Progress Update

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Changes since the last update (September 25, 2026):

- **All BLG analyses now cover the full 1000 ns for every replica.** Surface tension for R1/R2/R3, the last 500 ns gap, is closed; γ is unchanged qualitatively (51.9–52.3 mN/m, all four replicas).
- **Correction to previously reported CAS values.** The analysis selection had omitted the five phosphoserine residues (15, 17, 18, 19, 35); fixed and rerun. N-term SASA: CENTER 25.81 ± 2.39 (was 25.41 ± 1.97), R1 25.42 ± 1.45 (was 24.77 ± 1.40) nm^2. DSSP coil: 69.8% / 76.1% (was 71.0% / 76.6%). Contact unchanged (99.5%). Conclusions unchanged. BLG was not affected.
- **Accessible-feature SASA normalised for region size.** The ∼7-fold raw difference between CAS N-term (25 residues) and BLG calyx (9 residues) reported on September 25 is mostly a size effect. Per residue it is ∼2.7-fold, and as a fraction of each residue's own surface ∼2.6-fold (Table 2). CAS remains the more exposed, but the contrast is smaller than stated.
- **Figure 2 (structural dynamics) now complete for both proteins:** RMSD, RMSF and R_g, per replica.

BLG

Replica (1000 ns)	Calyx SASA (nm^2)	R_g (nm)	γ (mN/m)	Patch RMSD (nm)
CENTER	3.81 ± 0.46	1.504 ± 0.022	51.9 ± 0.5	0.198 ± 0.081
R1	3.15 ± 0.34	1.497 ± 0.009	52.1 ± 0.4	0.347 ± 0.088
R2	3.59 ± 0.32	1.494 ± 0.009	52.3 ± 0.2	0.254 ± 0.059
R3	3.36 ± 0.48	1.507 ± 0.014	51.9 ± 0.5	0.276 ± 0.064

Table 1: BLG per-replica summary, full 1000 ns each. γ is given as mean ± block-averaged SEM (10 blocks of 100 ns); the per-frame SD is ∼38 mN/m. For R1/R2/R3, γ was previously 52.7/52.0/52.6 mN/m from the first 500 ns only.

	Region (residues)	Raw (nm^2)	Per residue (nm^2)	Exposed fraction†
BLG calyx (mean of 4 replicas)	9	3.48	0.39	0.13
CAS N-term (mean of 2 replicas)	25	25.61	1.02	0.34
CAS / BLG		7.4	2.7	2.6

Table 2: Accessible-feature SASA, three normalisations. Computed on 401 frames (every 5 ns) per replica. †Region SASA in the full protein divided by the sum of each residue's SASA alone in the same frame. CAS values include the five phosphoserines.

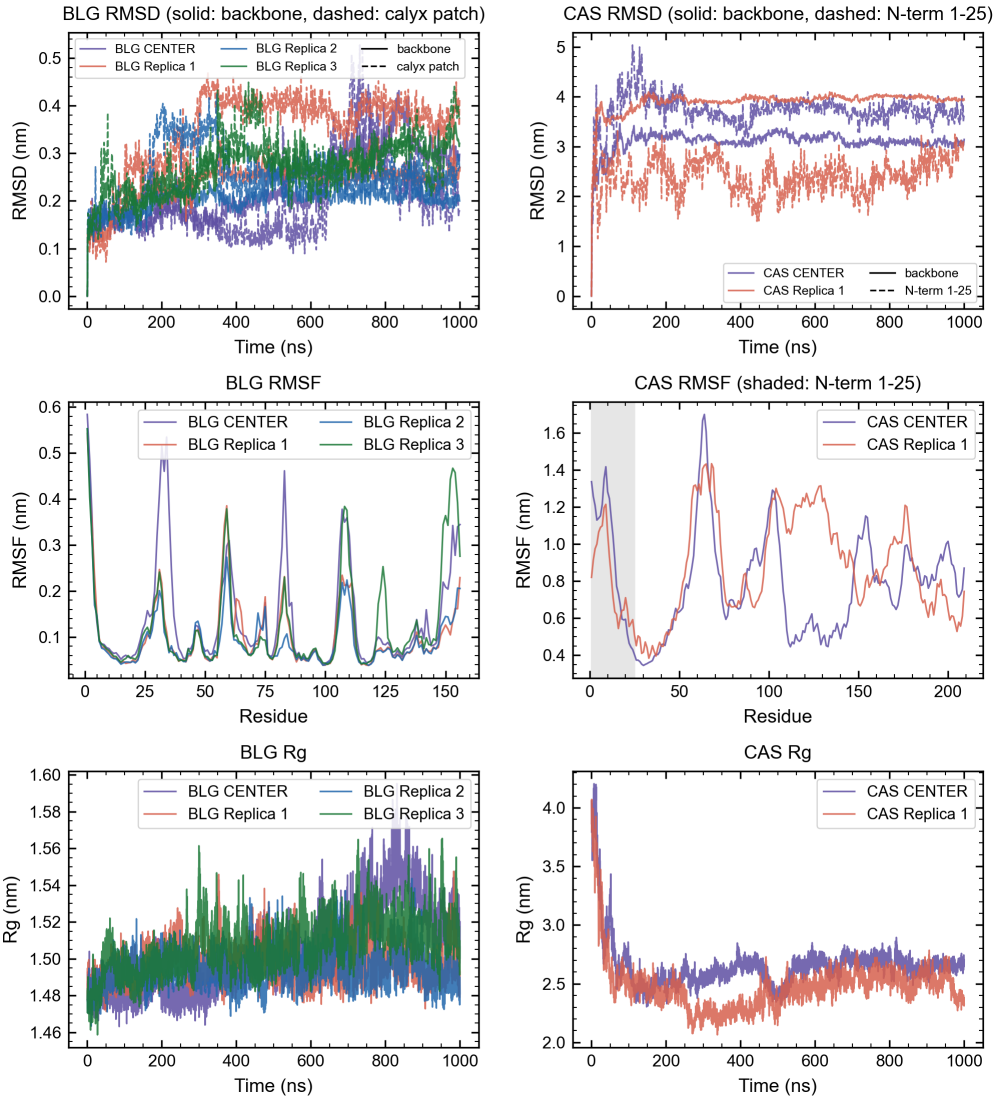


Figure 1: Structural dynamics, per replica (not averaged). Left: BLG. Right: CAS. Top: RMSD, backbone (solid) and calyx patch / N-term 1–25 (dashed). Middle: per-residue RMSF. Bottom: radius of gyration. Axes differ between proteins; CAS is disordered, so its RMSD/RMSF reflect chain mobility. R1 BLG's patch RMSD stays near 0.4 nm from ∼300 ns, after its longest contact event.